Kuma McCraw

kmccraw@college.harvard.edu • linkedin.com/in/kuma-mccraw/ • 510 239 8644 • Richmond, CA

Education

Harvard University — Cambridge, MA

B.S in Mechanical Engineering. GPA: 3.9/4.0 — May 2026

Relevant Coursework: Computer-Aided Machine Design, Fluid Mechanics, Thermodynamics, Mechanical Systems, Mechanics of Solids, Computing with Python, Materials, Intermediate Chinese, Statistical Inference, AI

The Thacher School — Ojai, CA

GPA: 4.7/4.0; SAT: 1510(800 math); 1st Place FTC Robotics Collins Innovate Award; Thacher Cup — June 2022

Experience

Apple — Cupertino, CA

Hardware Engineering Intern - Camera Module Reliability — May 2025 – Aug 2025

- Designed, fabricated, and implemented 3 new multi-part fixtures for module and coupon-level testing
- Developed ML-based generative design model for component level topology optimization and validation
- Developed quasi-static FEA model to simulate component level stress on modified components I designed
- Analyzed data with JMP (for Weibull analysis) and Python, uncovering insights that drove critical design changes and demonstrated strong correlation with both computational predictions and system-level failure modes & rates

Ford Motor Company — Dearborn, MI

Powertrain Design Engineering Intern — May 2024 – Aug 2024

- Utilized CATIA to design new power steering module mounting bracket for future trucks
- Implemented critical porosity reduction and cost-saving design changes for 3.0L engine covers
- Redesigned 6.8L engine water outlets to streamline supplier manufacturing
- Updated BOMS & impacted engineering drawings for cylinder heads across 3 vehicle programs
- Tested beta dashboard software features for functionality and user experience

Harvard Ability Lab — Cambridge, MA

Assistive Robotics Researcher — Sep 2023 – May 2024

- Designed a modular, haptic feedback enabled augmented cane attachment to enhance obstacle avoidance for the visually impaired using SolidWorks
- Fabricated and assembled laser cut and 3D components to deliver rapid prototypes to the PI

PricewaterhouseCoopers (PwC) — Shanghai, China

Environmental, Social, and Governance (ESG) Summer Associate — June 2023 – Aug 2023

- Developed a sustainable design framework for a $100M real estate project to achieve LEED certification

Projects & Leadership

3D-printed drone – designed, fabricated, and soldered a 500g Drone with a 6:1 thrust to weight ratio
Autonomous plant-care system – designed, 3D-printed and programmed a self-watering ecosystem for house plants using an Arduino, soil moisture sensor, neo-pixels, liquid crystal display, and 3V submersible pump
Dexterous All-Terrain Robot – Worked with team to design and machine all components - I was responsible for the belt & pulley driven arm & claw mechanism (1st place in Harvard Undergraduate ES51 Robotics Competition 2023)

National Society of Black Engineers – Harvard Chapter Engineering Representative (2023-2024)
Harvard World Model United Nations – Chair of Disarmament & International Security Committee (2024), Under-Secretary General for Business (2025)

Skills

Technical: SolidWorks [Certified], CATIA, NX, COMSOL, GD&T, DFM, PLM, Python, MATLAB, SQL, Arduino, JMP
Engineering Portfolio: kumamccraw.muse.io/